

Thermal, mechanical and molecular relaxation properties of stabilized sucrose solutions at sub-zero temperatures

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Sub-zero behaviour of sucrose solutions in the presence of xanthan gum, gelatin and guar gum were investigated using differential scanning calorimetry (DSC), thermomechanical analysis (TMA) and ^1H nuclear magnetic resonance (NMR). Two freezing rate and thermal history dependent phase transitions, denoted T_{Tr1} and T_{Tr2} , were observed. Only properly annealed xanthan systems had any influence on the thermal properties of freeze-concentrated sucrose solutions (DSC). However, stabilizers altered the rate of thermal deformation (decrease), calculated viscosity (increase) and compliance (decrease) above T_{Tr2} , xanthan having the largest effect. These properties appeared to be enhanced through slow freezing consistent with changes in free volume. Also, the presence of guar gum appeared to change the NMR t_2 relaxation properties of the system. It is postulated that a mechanism of stabilizer action at sub-zero temperatures involves a modification of the kinetic properties of the unfrozen phase rather than thermodynamic parameters.

Keywords: Stabilizers, glass transition, sub-zero temperatures.

INTRODUCTION

Food materials stored at sub-zero temperatures contain a significant unfrozen phase (UFP) due to the process of freeze-concentration (Goff, 1992). This UFP can participate in or support several deleterious reactions such as structural collapse, crystal growth and enzymatic activity, consequently reducing the overall textural and quality attributes of the product. Control of ice crystal growth in products such as ice cream, especially during temperature-abusive conditions, is commonly achieved by the addition of high molecular weight polysaccharide stabilizers which act to control the growth of ice crystals (Caldwell *et al.*, 1992). Complex stabilizers also enhance body and provide stiffness during freezing, ensuring acceptable sensory qualities to frozen products. However, researchers examining the role of stabilizers in freeze-concentrated systems have found no consistent evidence to suggest stabilizers alter the ice nucleation process, linear crystallization rates or

the phase equilibrium of frozen systems (Muhr & Blanchard, 1986; Muhr *et al.*, 1986; Budiawan & Fennema, 1987a; Buyong & Fennema, 1988). In addition, an attempt to correlate initial mix viscosity to stabilizer action has been inconclusive (Budiawan & Fennema, 1987b). It has been hypothesized that once the critical concentration of the polymer is reached, due to freeze-concentration, intense overlap and entanglement may have a dramatic effect on the diffusion kinetics and molecular response of the viscoelastic UFP surrounding the ice crystals (Goff *et al.*, 1993).

Much of the recent research into food stability has focused on the kinetics and formation of a glassy state during the freeze-concentration process (Simatos & Blond, 1991; Slade & Levine, 1991; Roos & Karel, 1991a,b). In freeze-concentrated sucrose systems, the UFP reverts into a disordered amorphous (glassy) state when the sucrose concentration reaches approximately 80%. This glassy domain is relatively unstable and exhibits solid-like characteristics and high viscosity, $>10^{13}$ Pa s. Constituents within the UFP would exhibit an almost complete reduction in translational mobility

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and diffusion properties, hence becoming kinetically immobilized and unable to support or participate in reactions (i.e. enzymatic activity). However, phase behaviour in frozen sucrose solutions is relatively complex resulting in two highly associated thermal transitions representative of a glass-rubber transition. Onset of glass relaxation is currently thought to occur at the lower temperature transition while the warmer second transition represents the onset of ice melting-dissolution coupled with a rearrangement process (Roos & Karel, 1991a,b; Ablett *et al.*, 1992; Sahagian & Goff, 1994).

Phase behaviour of frozen systems is commonly characterized using differential scanning calorimetry (DSC), thermomechanical analysis (TMA) and dynamic mechanical analysis (Slade & Levine, 1991; LeMeste & Huang, 1992). Nuclear magnetic resonance (NMR) is a non-invasive technique which has been used extensively to evaluate water mobility and dynamics in sucrose and food systems at ambient temperature (Hills *et al.*, 1990; Padua & Schmidt, 1992). However, only a limited number of NMR studies focusing on the crystallization process (Weisser & Harz, 1984) or second-order phase behaviour (Rubin *et al.*, 1990) at sub-zero temperatures have been performed. Determination of relaxation properties of protons dispersed within the unfrozen phase, from measurement of the ^1H spin-spin (t_2) relaxation time, may provide insight into the molecular mobility and nature of the system. The overall objective of this study was to further describe the influence of stabilizers on various sub-zero thermal events in sucrose solutions using the above techniques.

METHODS AND MATERIALS

Preparation of solutions

Sucrose solutions at 20% (w/w) (Fisher Scientific Co. Ltd, Whitby, ON) were prepared in double distilled deionized water alone and in the presence of 0.5% (w/w) xanthan gum (Sigma Chemical Co. St Louis, MO), gelatin (275 bloom), (Fisher Scientific Co. Ltd, Whitby, ON) and guar gum (Food Specialties, Mississauga, ON). Sucrose-stabilizer blends were slowly added to 65°C water, vigorously mixed into solution and held at 75°C for 10 min. Moisture driven off during preparation was added back. Solutions were gradually cooled to 4°C and aged for 12 hr before being analysed.

Thermal behaviour—differential scanning calorimetry (DSC)

Thermal characteristics of the various solutions were determined using a DuPont Thermal Analyst 2000 system (TA Instruments, New Castle, DE) equipped with a 2910 cell base according to previously described methods (Sahagian & Goff, 1994). Temperature cali-

bration was determined using gallium, indium and ethylene glycol (Sigma Chemical Co., St Louis, MO) as standards. Cell constant (heat flow) calibration was determined using gallium. Annealed samples were slowly frozen, warmed to -35°C , held isothermally for 60 min then cooled to -80°C at $30^\circ\text{C min}^{-1}$. Transition temperatures T_{Tr1} and T_{Tr2} and changes in specific heat capacity (ΔC_p) were calculated according to previously described methods (Sahagian & Goff, 1994).

Mechanical behaviour—thermomechanical analysis (TMA)

Changes in dimension or specific volume as a function of temperature were performed on a DuPont 2940 module (TA Instruments, New Castle, DE) using the parallel plate rheometer attachment. Sample preparation and method of analysis were performed according to previously described methods (Sahagian & Goff, 1994). Analysis was performed under a constant nitrogen flush. Apparent viscosity as a function of temperature was calculated from the data points of the TMA curve using the following equation:

$$\eta = \frac{4F}{\frac{3\pi r^4}{\Delta l/h^2} \Delta t} \quad (1)$$

where r is the sample radius (4.75×10^{-3} m), h is the sample height (4.95×10^{-4} m), F is the applied force (in Newtons) and Δt is the change in time (Du Pont Instruments, 1981). The upper viscosity limit of this technique with certainty is 10^8 Pa s. Viscosities were calculated starting from the first change in height after T_{Tr2} .

Proton (^1H) nuclear magnetic resonance (NMR)

Proton (^1H) NMR measurements were performed on sucrose solutions alone and in the presence of 0.5% (w/w) guar gum using a Varian Unity 400 Spectrometer equipped with a variable temperature liquid nitrogen vapour control unit and a 9.4 T magnet. Samples (1 ml) in 5 mm NMR tubes were placed into the NMR probe and cooled at an approximate rate of 5°C min^{-1} . At several points between -60 and -10°C the samples and probe were allowed to equilibrate for 5 min before the probe was tuned and the 90° pulse width was determined for both systems. Spectra were obtained using a spectral width of 50 000 Hz from a single $[\tau_1-90^\circ\text{-acquire}]_n$ pulse sequence where τ_1 is the delay time between acquisitions (five times the T_1). A single resonance peak was observed at temperatures below -5°C . Samples were scanned 100 times before the FID was Fourier transformed using a line broadening function of 150 Hz. Transverse relaxation times (t_2 , ms) at all temperatures were calculated by measuring the line

width at half height of the NMR spectra using the equation:

$$t_2 = \frac{1}{(\pi\Delta\nu)} \quad (2)$$

where $\Delta\nu$ is the line width at half height of the NMR spectrum. As opposed to NMR convention, transverse relaxation times are denoted as t_2 throughout this paper to differentiate time from temperature (T).

Statistical analyses

Data were analysed by the Analysis of Variance and Tukey's test procedures of SAS (Statistical Analysis Systems, Cary, NC). Decisions were made at the 5% ($P \leq 0.05$) significance level from four and six replicates in the DSC study and four in the TMA study.

RESULTS AND DISCUSSION

Thermodynamic properties of rapidly-frozen solutions

Freezing of stabilized sucrose solutions led to the formation of two highly associated transitions, identified as T_{Tr1} and T_{Tr2} . Thermal properties of sucrose solutions in

the presence of xanthan gum, gelatin and guar gum were found to be stabilizer and freezing rate dependent (Tables 1 and 2). Under rapid freezing, the glass transition onset (denoted T_{Tr1}) for 20% (w/w) sucrose solutions was -54.6°C while the xanthan gum system showed a lower T_{Tr1} of -58.1°C (Table 1). Gelatin and guar gum appeared to have no effect on the experimental T_{Tr1} onset. It has been shown that the potential for partial freeze-concentration increases with initial sucrose concentration and freezing rate, resulting in a decreased T_g (Sahagian & Goff, 1994). Roos & Karel (1991a,b) have also shown that excess plasticizer resulting from delayed crystallization will lower T_g of non-annealed systems. Therefore, it is possible that the presence of xanthan within the UFP initially altered the kinetics of ice crystallization during rapid freezing. Alternatively, xanthan molecules may have increased the volume and enthalpy defects of the glass causing a reduction in T_{Tr1} (characteristic of T_g). Change in heat capacity (ΔC_p) associated with this transition was also lower in the xanthan systems (Table 1). This suggests that the presence of xanthan gum causes a fundamental change in the structural makeup of the amorphous state.

After rapid freezing, a measurable exothermic peak (ΔH_{ex}) between T_{Tr1} and T_{Tr2} during warming suggested that supercooled water trapped in the sucrose glass

Table 1. Thermodynamic data for 20% (w/w) sucrose alone and in the presence of 0.5% (w/w) xanthan gum, gelatin (275 bloom) and guar gum rapidly frozen and warmed at 5°C min^{-1} determined by DSC^a

Sample % (w/w)	T_{Tr1on}^b ($^\circ\text{C}$)	$\Delta C_p T_{Tr1}^c$ ($\text{J g}^{-1} ^\circ\text{C}^{-1}$)	ΔH_{ex}^d (J/g)	T_{Tr2on}^e ($^\circ\text{C}$)	T_{Tr2mid}^f ($^\circ\text{C}$)	$\Delta C_p T_{Tr2}^g$ ($\text{J g}^{-1} ^\circ\text{C}^{-1}$)	ΔH_m^g (J/g)
Sucrose	$-54.6ab^h$	0.75b	1.64b	$-36.3ab$	$-32.2b$	3.4b	213.4a
Sucrose/xanthan	$-58.1a$	0.48a	1.93a	$-38.6a$	$-34.7a$	2.6c	215.4a
Sucrose/gelatin	$-53.9b$	0.84c	0.92c	$-36.8ab$	$-32.7b$	3.6ab	218.9a
Sucrose/guar	$-54.2b$	0.85c	1.72ab	$-35.6b$	$-32.5b$	3.8a	217.4a

^a $n = 6$.

^b Onset of T_{Tr1} .

^c Per gram of sucrose.

^d Enthalpy of devitrification peak after T_{Tr1} .

^e Onset of T_{Tr2} .

^f Midpoint of T_{Tr2} .

^g Enthalpy of melt per gram of solution.

Table 2. Thermodynamic data for 20% (w/w) sucrose with and without 0.5% (w/w) xanthan gum, gelatin (275 bloom) and guar gum cooled at 2°C min^{-1} , annealed at -35°C for 60 min and warmed at 5°C min^{-1} , as measured by DSC^a

Sample % (w/w)	T_{Tr1on}^b ($^\circ\text{C}$)	$\Delta C_p T_{Tr1}^c$ ($\text{J g}^{-1} ^\circ\text{C}^{-1}$)	T_{Tr2mid}^d ($^\circ\text{C}$)	$\Delta C_p T_{Tr2}^c$ ($\text{J g}^{-1} ^\circ\text{C}^{-1}$)	ΔH_m^e (J g^{-1})
Sucrose	$-40.5a^f$	1.23a	$-31.4a$	3.81a	220.2a
Sucrose/xanthan	$-42.5a$	1.31a	$-32.2a$	3.12b	219.1a
Sucrose/gelatin	$-42.8a$	1.22a	$-32.6a$	3.75a	206.2b
Sucrose/guar	$-42.5a$	1.24a	$-31.4a$	3.76a	222.3a

^a $n = 4$.

^b Onset of T_{Tr1} .

^c Per gram of sucrose.

^d Midpoint of T_{Tr2} .

^e Enthalpy of melt per gram of solution.

^f Means followed by the same letter within each column are not significantly different ($P > 0.05$).

regained appreciable translational mobility to spontaneously crystallize within the time frame of the experiment ($5^{\circ}\text{C min}^{-1}$). In the Food Science and cryobiology literature this process is called devitrification. The presence of this measurable devitrification exotherm demonstrates the large effect kinetic properties can have on a thermodynamic driven event (crystallization process), hence, increasing the potential for non-equilibrium freezing. Sucrose showed a devitrification peak of 1.6 J g^{-1} while the other stabilizers, except gelatin, exhibited similar magnitudes of crystallization (Table 1). This result suggests the gelatin system may have been closer to maximum freeze-concentration at T_{Tr1} since the estimated total ice content at T_{Tr2} (ΔH_m) was similar to the other systems.

Transition temperatures related to the T_{Tr2} followed the same trend as observed for the T_{Tr1} . The presence of xanthan gum appeared to shift the T_{Tr2} onset temperature about 2°C lower (-38.6°C) than the value observed for sucrose alone (-36.3°C). ΔC_p values for this transition were reduced by the presence of xanthan gum (Table 1). This could be the result of two phenomena. The energy required for this transition may have been reduced through a change in the structure of the system and/or the xanthan polymers physically hindered the relaxation process, within the time frame of the measurement. However, in general, rapidly frozen systems were unstable and difficult to reproduce. T_{Tr1} onset values were $7\text{--}15^{\circ}\text{C}$ lower than published T_g' values for sucrose (Slade & Levine, 1991; Ablett *et al.*, 1992; Ross & Karel, 1991a,b; Sahagian & Goff, 1994) suggesting these systems were far from structural equilibrium (crystalline and amorphous).

Thermodynamic properties of annealed solutions

Systems slowly frozen showed little or no measurable ΔH_{ex} and T_{Tr1on} values up to 8°C higher than observed after rapid freezing which is consistent with the T_g' proposed by Roos and Karel (1991a,b). Although devitrification was seldom observed, separation of the two transitions of approximately 5°C was always evident, and therefore, considered an indication that the systems were not maximally freeze-concentrated. In an attempt to facilitate maximum ice content and stability after slow freezing, samples were annealed according to a specific optimal protocol (Sahagian & Goff, 1994). The significance of adequate annealing has been clearly demonstrated in recent studies of frozen sugar solutions (Roos & Karel, 1991a,b; Wasylyk & Baust, 1991; Ablett *et al.*, 1992; Sahagian & Goff, 1994). The presence of stabilizers in slowly frozen-annealed systems did not significantly ($P > 0.05$) influence either relaxation temperature (T_{Tr1on} , -41°C or T_{Tr2mid} , -32°C) and the two transitions appeared to be highly associated but separate events (Table 2). The T_{Tr1} value for sucrose alone was considered the T_g' of the maximally

freeze-concentrated sucrose phase. Slightly lower values for the stabilized systems may indicate that maximum freeze-concentration still had not occurred.

The glass transition is a function of the mean molecular weight and related free volume distribution (Turnbull & Cohen, 1961; Ferry, 1980; Tant & Wilkes, 1981; Slade & Levine, 1991). The data suggested that after annealing, the presence of these highly concentrated stabilizers within the glassy matrix was not significant ($P > 0.05$) to modify the thermodynamic properties (e.g. T_g') of the glass (Table 2). Heat capacity change at T_{Tr2} , however, was significantly ($P \leq 0.05$) altered in the presence of xanthan gum (Table 2). A lower ΔC_p at T_{Tr2} was also observed after rapid and slow freezing (Tables 1 and 2). This relaxation may involve both first- and second-order processes with emphasis on the latter (Slade & Levine, 1991). However, Luyet and Rasmussen (1968), and recently, Roos and Karel (1991a,b) believe this transition represents only the onset of ice melting, while Wasylyk and Baust (1991) have suggested it involves differences in ordering of hydration molecules. Independent of freezing conditions, it appears that interactions between rod-like segments of xanthan have interfered with the ice dissolution process and/or the relaxation of water molecules associated with the sucrose concentrated UFP. Since the T_{Tr2} relaxation is much larger than the T_{Tr1} , any change in the apparent magnitude may relate to a modification of the overall kinetics of the system.

Thermomechanical behaviour of stabilized sucrose solutions

Stabilized systems after rapid freezing showed varied levels of volumetric collapse during warming through the vitreous state, $T < T_{\text{Tr1}}$ (data not shown). Volume reduction appeared to be enhanced by the presence of gelatin while xanthan prevented rearrangement and mechanical collapse. Amorphous materials exhibit a strong potential to undergo rearrangement mechanisms in order to reduce excess volume and enthalpy after rapid cooling (Tant & Wilkes, 1981). The process is a function of the translational mobility and free volume size distribution of the glass. This observation suggests that xanthan takes on a conformation which manifests different kinetic factors hindering rearrangement by altering free volume. This phenomenon may be related to the apparent reduction in ΔC_p (0.48 J g^{-1}) at T_{Tr1} induced by the presence of this polymer (Table 1).

Stabilized systems slowly frozen and annealed at -35°C for 30 min in the TMA showed different behaviour. During rewarming, a distinct change in the slope (expansion coefficient) of the TMA curve around -47°C was observed for all solutions (Table 3) demonstrating an increase in free volume at T_g . This increase in volume after annealing of amorphous frozen sucrose solutions and aged polymers has also been observed elsewhere (Jo & Ko, 1991; LeMeste & Huang, 1992; Sahagian & Goff, 1994). Based on this interpretation,

Table 3. T_{Tr1} , T_{Tr2} , calculated compliance (-26°C) and apparent viscosity (-15°C) for 20% (w/w) sucrose with and without 0.5% (w/w) xanthan gum, guar gum and gelatin (275 bloom) under a constant stress of 140 Pa, measured by TMA^a

Sample % (w/w)	Slow freeze ^{b,c} annealed -35°C for 30 min		Slow freezing ^b		Rapid freezing
	T_{Tr1} ($^{\circ}\text{C}$)	T_{Tr2} ($^{\circ}\text{C}$)	η^d (Pa s) $\times 10^6$	Compliance (Pa^{-1}) $\times 10^{-5}$	Compliance (Pa^{-1}) $\times 10^{-5}$
Sucrose	-47.3a ^e	-30.8a	1.18c	5.38a	27.7a
Sucrose/xanthan	-48.8a	-28.5a	2.33a	3.17b	3.45c
Sucrose/gelatin	-48.1a	-28.2a	1.65b	4.14ab	17.9b
Sucrose/guar	-46.9a	-29.1a	1.55b	3.98ab	15.9b

^a $n = 4$.^b Approximate freezing rate of $2^{\circ}\text{C min}^{-1}$.^c Transition temperatures for non-annealed were similar.^d Wall shear rate of $2.11 \times 10^{-3} \text{ s}^{-1}$.^e Means followed by the same letter within each column are not significantly different ($P > 0.05$).

stabilizers had no significant effect ($P > 0.05$) on the mechanical T_{Tr1} onset (Table 3). This trend correlated with the DSC measurements, however, the absolute values were much lower. This indicates that with the sample geometry used in TMA, longer time frames were required to initiate maximum freeze-concentration. Onset of T_{Tr2} (Table 3) represented as a collapse process was found to correlate reasonably with T_{Tr2} values from DSC (Table 2). Stabilizers did not significantly influence this relaxation temperature, however, but did increase the onset by a couple of degrees: sucrose -30.8°C , sucrose-xanthan -28.5°C (Table 3).

The influence of stabilizers on structural flow and collapse above T_{Tr2} was substantial. Following rapid freezing, a strong resistance to thermal deformation was observed for all stabilizers, xanthan gum having the largest effect (Fig. 1). Goff *et al.* (1993) have also

shown this type of behaviour with frozen sucrose-guar solutions above the T_g' . However, under the conditions of this study, it is evident that the presence of xanthan gum and gelatin increased resistance to structural rearrangement and viscous flow to a greater amount than guar gum. The data suggest that molecular structure of the polymer plays an important role in mechanical stability at temperatures greater than T_{Tr2} (Fig. 1).

Calculated compliance values show the large influence that stabilizers and freezing rate had on structural integrity of frozen systems at -26°C (Table 3). After rapid freezing, all stabilizers had a significant influence ($P \leq 0.05$), reducing the compliance by approximately 42% for guar gum and 35% for gelatin. Xanthan showed an eight-fold decrease in compliance and was significantly different from the other two stabilizers (Table 3). In fact, the effect of slow freezing appeared to substantially reduce compliance values, compared to rapid freezing, consistent with an overall reduction in free volume (Tant & Wilkes, 1981). The presence of xanthan in slowly frozen sucrose systems produced a significant reduction ($P \leq 0.05$) in compliance compared to sucrose alone; however, when compared to each other, no significant effect ($P > 0.05$) was seen between stabilizers under these conditions (Table 3). It appears that the relaxation occurring within the T_{Tr2} temperature range was affected by the presence of xanthan gum, since ΔC_p at T_{Tr2} (Table 2) and compliance values were both reduced compared to sucrose alone. Xanthan gum appeared to promote the formation of a more stable system directly affecting mechanical properties (Fig. 1).

Calculated viscosity at -15°C for each system slowly frozen is shown in Table 3. At this temperature a large increase in molecular diffusion and flow is expected as free volume increase (excess plasticizer) and ice dissolution (dilution) occur concurrently (Levine & Slade, 1990; Slade & Levine, 1991; Simatos & Blond, 1991).

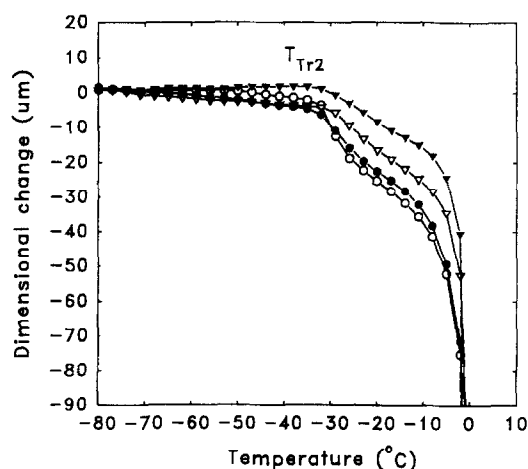


Fig. 1. Influence of gelatin (∇), xanthan gum ($\blacktriangledown$) and guar gum ($\bullet$) (0.5% w/w) on the rate of mechanical deformation in rapidly frozen (liquid nitrogen) 20% (w/w) sucrose solutions ($\circ$) in the post T_{Tr2} temperature range. After rapid freezing systems were equilibrated at -80°C for 3 min and warmed at a rate of $2^{\circ}\text{C min}^{-1}$ under a stress of 140 Pa. Data points represent the mean of three determinations.

All three stabilizers significantly increased ($P \leq 0.05$) calculated viscosity; however, xanthan was found to have a significantly greater effect ($P \leq 0.05$) than guar or gelatin (Table 3). The presence of xanthan gum caused an approximate two-fold increase in viscosity (2.3×10^6 Pa s) compared to sucrose alone (1.2×10^6 Pa s). This translates into a substantial reduction in diffusion kinetics and mobility of the rubbery unfrozen component. Viscosity profiles also suggest that stabilizer behaviour at sub-zero temperatures is structure dependent.

Stabilizer influence on spin-spin relaxation time (t_2)

Proton (^1H) NMR measurements within the glassy and rubbery regions were performed to determine the effect of the glass transition and guar gum on proton rotational mobility. Relaxation times, t_2 , determined from the line width of both systems are presented in Figure 2. Transverse relaxation times are a function of chemical composition, mobility, temperature and structure (Schrader *et al.*, 1992). During t_2 transverse or spin-spin relaxation, loss of phase coherence in the x' y' plane after a 90° pulse is facilitated by an exchange of energy between nuclei. Increases in t_2 are generally related to an increase in the rotational motion of a molecule leading to a narrowing of the resonance line width (Hemminga, 1992). A slight increase in t_2 was seen around -38°C followed by a large change starting at approximately -30°C (Fig. 2). Using ESR, Roozen and Hemminga (1990) have reported a drastic decrease in τ_c for a frozen 20% sucrose solution at approximately -32°C representing an increase in molecular mobility of the system. The change in t_2 at $T_{\text{Tr}2}$ in this study correlates well with their results. These two apparent increases in t_2 correlated well with the $T_{\text{Tr}1}$

and $T_{\text{Tr}2}$ observed from thermal analysis. In addition, the relative change in magnitude for both transitions appears to be in agreement with the DSC ΔC_p measurements.

Rubin *et al.* (1990) reported a correlation between the temperature corresponding to a reduction in peak amplitude of the FID and the glass transition of vitreous honey at sub-zero temperatures. Increase in t_2 with the 20% sucrose solution may, in fact, represent a change in proton mobility at T_g ($T_{\text{Tr}1}$ from DSC). Sucrose solutions in the presence of guar gum (0.5% w/w) showed relatively smaller t_2 values between the transition regions $T_{\text{Tr}1}$ and $T_{\text{Tr}2}$ (Fig. 2). Assuming this change is related to T_g , viscosity would decrease several orders of magnitude thereby facilitating rearrangement and an overall change in the kinetics of the system. Although this reduction in t_2 is small, ^1H NMR results suggest that one or perhaps several kinetic variables were modified in the presence of the polymer before phase change occurred. Alternatively, proton relaxation mechanisms (related to rotational mobility) may have been altered possibly through a structuring-ordering effect of the water being sampled (detectable).

At temperatures above $T_{\text{Tr}2}$, guar appeared to reduce t_2 relaxation times similar to trends observed with the stabilized TMA plots (Fig. 1). For example, at -23°C , t_2 of the stabilized system was on average approximately 20% smaller than the corresponding frozen system without stabilizer. It is possible that the polymer influenced proton exchange rates between different sites such that broadening effects occurred (Lai & Schmidt, 1991). Specifically, this modification in proton behaviour may stem from an increase in the ratio between the motionally restricted (short t_2) and the bulk (water) protons resulting in a weighted-average reduction in t_2 (Cole *et al.*, 1993). It is worth noting that based on equilibrium thermodynamics (freezing curve) the quantity of UFW increases with temperature above $T_{\text{Tr}2}$. Therefore, in the case of sucrose alone, the observed increase in t_2 most likely represents a combined effect of decreased viscosity (dilution) and increased proportion of protons available for detection. However, since the guar gum system did not significantly influence the relative quantity of ice as estimated by DSC (Tables 1 and 2), it is possible that the effect demonstrated in Figure 2 evolves from changes in the self-diffusion properties of the NMR detectable protons dispersed within the polymer. In addition, an increase in intrinsic viscosity due to entangled guar gum polymers may have led to a reduction in proton mobility and mass transfer properties (Tait *et al.*, 1972; Padua & Schmidt, 1992); however, this interpretation has been strongly criticized (Mora-Gutierrez & Baianu, 1989). The latter authors believe relaxation is simply governed by exchange between bulk and bound water, not by the size and shape of the macromolecule and resulting resistance to water flow under shear stress (viscosity). At

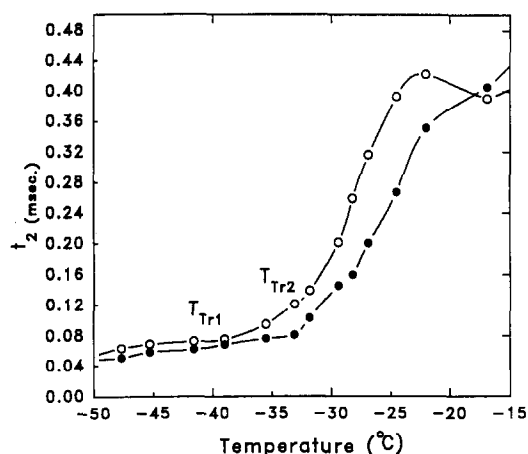


Fig. 2. NMR transverse relaxation time (t_2) at various sub-zero temperatures for slowly frozen (approximately 5°C min^{-1}) 20% (w/w) sucrose solutions (○) alone and in the presence of 0.5% (w/w) guar gum (●). t_2 was calculated from Equation (2) using the line width analysis of the single resonance determined from the spectrometer software. Data points represent the mean of two determinations.

present, a full explanation for this effect on t_2 is not available and further investigation is needed to test the above interpretations.

CONCLUSIONS

The results of this study suggest that stabilizer action involves a modification of the transport properties of the UFP. TMA data have revealed that volumetric relaxation during warming of the glassy region and resistance to structural flow past T_{Tr2} (viscosity) may be polymer dependent. This suggests that rod-like interactions (xanthan) and stabilizers which form true junction zones (gelatin) are more stable (rapidly reformed) and hence, interfere with rearrangement mechanisms within both physical states (amorphous and rubbery). As the concentration of the UFP reaches the semi-dilute region of the polymer, side chain and main backbone non-specific entanglements start to occur (Morris *et al.*, 1981). Xanthan molecules may act more rigidly once the critical concentration is reached and, therefore, may experience difficulties in aligning with the shear field. It is possible that a reduction in free volume distribution by the presence of the polymers may have severely reduced diffusion of the polymer itself, which in turn, reduces the overall kinetics (translational mobility) of the unfrozen phase (LeMeste *et al.*, 1991). As reflected in a reduction in t_2 past T_{Tr2} , NMR has shown that the addition of a polymer at sub-zero temperatures changes the proton relaxation properties of the system. It is hypothesized that stabilizers may change the ordering of the unfrozen water molecules and resulting kinetics, such that recrystallization is encouraged to occur within the dispersed polymer, rather than existing crystals.

ACKNOWLEDGEMENTS

The authors wish to thank B. Cheeseman for NMR technical support and the Natural Sciences and Engineering Research Council of Canada and Ontario Ministry of Agriculture and Food for financial support of this project.

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(Received 6 December 1993; revised version accepted 23 April 1994)